

Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 2: Arithmetic Expressions

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	2 — Arithmetic Expressions
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. the deadliest traps are unflipped signs when a minus sits before a nested bracket, premature left-to-right evaluation of $\times$ and $\div$, half-distribution that multiplies only the first inner term, and assuming subtraction or division can be regrouped like addition and multiplication. Do not write on this paper — mark answers on your answer sheet.

1. Evaluate: $300 - 5 \times (24 - 2 \times (9 - 2 \times 3)) + 7$.

(A) 307

(B) 217

(C) 289

(D) 247

2. Which removal of brackets is CORRECT for $250 - (40 - 15 + 8 - 6 + 3)$?

(A) $250 - 40 + 15 - 8 + 6 - 3$

(B) $250 - 40 - 15 + 8 - 6 + 3$

(C) $250 - 40 + 15 + 8 + 6 + 3$

(D) $250 + 40 - 15 + 8 - 6 + 3$

3. Without computing either product, which is greater: 84×96 or $84 \times 100 - 84 \times 4$?

(A) 84×96

(B) They are equal

(C) $84 \times 100 - 84 \times 4$

(D) Cannot be decided

4. Evaluate: $720 \div (2 \times 3) \div 4 \times 5$.

(A) 6

(B) 24

(C) 150

(D) 9600

5. Written as a sum of signed terms, the expression $100 - 6 \times 4 + 72 \div 8 - 5 \times 2$ has which terms?

(A) 100, -6, 4, 72, 8, -5, 2

(B) 100, 24, 9, 10

(C) 100, -24, 9, -10

(D) 100, -24, -9, -10

6. Which expression equals $17 \times 100 - 17 \times 4 + 17$?

(A) 17×96

(B) 17×95

(C) 17×104

(D) 17×97

7. Without subtracting, which is greater: $613 - 287$ or $615 - 290$?

(A) $615 - 290$

(B) They are equal

(C) Cannot be decided

(D) $613 - 287$

8. Evaluate: $5 \times (40 - (12 + 2 \times (7 - 4))) - 18$.

(A) 110

(B) 82

(C) 92

(D) 182

9. Which single bracketed expression equals $120 - 18 - 24 - 11$?

(A) $120 - (18 + 24 + 11)$

(B) $120 - (18 - 24 - 11)$

(C) $120 - (18 + 24 - 11)$

(D) $(120 - 18) - (24 - 11)$

10. Without computing, which is greater: 49×51 or 50×50 ?

(A) 49×51

(B) 50×50

(C) They are equal

(D) Cannot be decided

11. Evaluate: $144 \div (36 \div (8 - 5)) + 6 \times 2$.

(A) 12

(B) 24

(C) 60

(D) 18

12. A worker is paid ₹95 per hour and a ₹120 daily allowance, but ₹15 per hour is deducted for canteen use, over an 8-hour day. Which expression gives the take-home pay in ₹?

(A) $8 \times (95 - 15) + 120$

(B) $8 \times 95 - 15 + 120$

(C) $8 \times (95 - 15 + 120)$

(D) $(8 \times 95 + 120) - 15$

13. Which statement about $200 - (60 - (25 - 8))$ is TRUE?

(A) It equals $200 - 60 - 25 - 8 = 107$

(B) It equals $200 - 60 + 25 + 8 = 173$

(C) It equals $200 - 60 + 25 - 8 = 157$

(D) It equals $200 - 60 - 25 + 8 = 123$

14. Evaluate: $96 \div (2 + 2 \times 3) \times 5 - 3 \times (15 - 2 \times 6)$.

(A) 3

(B) 60

(C) 33

(D) 51

15. Using the distributive property to compute 47×102 , the correct split and value are:

(A) $47 \times 100 + 2 = 4702$

(B) $47 \times 100 + 47 \times 2 = 4794$

(C) $47 \times 100 + 47 \times 100 = 9400$

(D) $47 \times 100 - 47 \times 2 = 4606$

16. To add $23 + 58 + 77 + 42 + 100$ quickly by pairing into round numbers, the best grouping and total are:

- (A) $(23 + 58) + (77 + 42) + 100 = 300$, but not round
(B) $(23 + 42) + (58 + 77) + 100 = 300$ (C) addition order is fixed, cannot regroup
(D) $(23 + 77) + (58 + 42) + 100 = 300$

17. Without computing, which is greater: 38×62 or 37×63 ?

- (A) 37×63 (B) They are equal
(C) 38×62 (D) Cannot be decided

18. Evaluate: $4 \times ((50 - 2 \times (6 + 4)) \div 5) + 7$.

- (A) 27 (B) 31
(C) 47 (D) 37

19. Which pair of expressions is EQUAL?

- (A) $60 - 18 + 7$ and $60 - (18 + 7)$ (B) $72 \div 6 \div 2$ and $72 \div (6 \div 2)$
(C) $60 - 18 - 7$ and $60 - (18 - 7)$ (D) $60 - 18 - 7$ and $60 - (18 + 7)$

20. Using the distributive idea, the value of $23 \times 8 + 23 \times 5 - 23 \times 3$ is:

- (A) 230 (B) 368
(C) 138 (D) 92

21. Evaluate $9 \times 7 - 4 \times 6 + 5 \times 8 - 3 \times 9$ by first collapsing each product to a signed term:

- (A) 52 (B) 108
(C) -52 (D) 28

22. Where must a single pair of brackets go so that $36 - 12 \div 2 + 4$ equals 16?

- (A) $(36 - 12) \div 2 + 4$ (B) $36 - 12 \div (2 + 4)$
(C) $36 - (12 \div 2 + 4)$ (D) $(36 - 12 \div 2) + 4$

23. Without computing, which is greater: $124 + 89 + 211$ or $126 + 88 + 209$?

- (A) $126 + 88 + 209$ (B) They are equal
(C) Cannot be decided (D) $124 + 89 + 211$

24. Evaluate: $250 - 3 \times (50 - (18 + 2 \times (10 - 4)))$.

- (A) 234 (B) 210
(C) 190 (D) 166

25. A student wrote $90 - (20 - (8 - 3)) = 90 - 20 - 8 - 3 = 59$. The error and the correct value are:

- (A) nothing is wrong; the value is 59
(B) each bracket level must be removed with its own sign-flip; the value is $90 - 20 + 8 - 3 = 75$
(C) only the outer bracket flips signs; the value is $90 - 20 + 8 + 3 = 81$
(D) both brackets are ignored; the value is $90 - 20 - 8 + 3 = 65$

26. Which expression equals 26×98 ?

(A) $26 \times 100 - 26 \times 2$

(B) $26 \times 100 - 2$

(C) $26 \times 90 + 8$

(D) $26 \times 100 + 26 \times 2$

27. Remove the brackets correctly: $80 - (25 - 12) + (18 - 9) - (6 - 2)$.

(A) $80 - 25 + 12 + 18 - 9 - 6 - 2$

(B) $80 - 25 - 12 + 18 - 9 - 6 + 2$

(C) $80 - 25 + 12 + 18 - 9 - 6 + 2$

(D) $80 - 25 + 12 + 18 + 9 - 6 + 2$

28. Evaluate: $100 - 60 \div 5 + 7 \times 3 - 24 \div 8 \times 2$.

(A) 109

(B) 103

(C) 97

(D) 115

29. Without computing, which is greater: 53×47 or 54×46 ?

(A) 54×46

(B) They are equal

(C) Cannot be decided

(D) 53×47

30. A shop offers 12 boxes, each box holding 8 chocolates, but 3 chocolates in every box are damaged. Which expression gives the good chocolates, and its value?

(A) $12 \times 8 - 3 = 93$

(B) $12 \times (8 - 3) = 60$

(C) $(12 - 3) \times 8 = 72$

(D) $12 \times 8 - 3 \times 12$ read as $12 \times 8 - 3 = 93$

31. Which expression has the GREATEST value?

(A) $3 + 5 \times 4 - 1$

(B) $3 \times 5 + 4 \times 1$

(C) $(3 + 5) \times (4 - 1)$

(D) $3 \times 5 - 4 + 1$

32. Evaluate: $60 - [50 - (20 - (8 - 3))]$.

(A) 25

(B) 15

(C) 35

(D) 45

33. By the distributive property, $34 \times 45 + 34 \times 55 - 34 \times 100$ equals:

(A) 0

(B) 3400

(C) 6800

(D) 34

34. Which equality is TRUE?

(A) $180 - (50 - 20 + 10) = 180 - 50 + 20 - 10$

(B) $180 - (50 - 20 + 10) = 180 - 50 - 20 + 10$

(C) $180 - (50 - 20 + 10) = 180 - 50 + 20 + 10$

(D) $180 - (50 - 20 + 10) = 180 + 50 - 20 + 10$

35. Without computing, which is greater: $7 \times (30 - 5)$ or $7 \times 30 - 5$?

(A) $7 \times (30 - 5)$

(B) They are equal

(C) Cannot be decided

(D) $7 \times 30 - 5$

36. Evaluate: $3 \times (8 + 2 \times (5 - 1)) - 2 \times (10 - 3 \times (4 - 2))$.

- (A) 40
(C) 24
- (B) 56
(D) 16

37. Five teams each have 9 players and 2 coaches; then 3 players in total leave the tournament. Which expression gives the people remaining?

- (A) $5 \times 9 + 2 - 3$
(C) $5 \times 9 + 2 \times 5 - 3 \times 5$
- (B) $5 \times (9 + 2 - 3)$
(D) $5 \times (9 + 2) - 3$

38. Which expression equals $19 \times 23 + 19 \times 17$?

- (A) $19 \times 23 + 17$
(C) 19×391
- (B) 38×40
(D) 19×40

39. Evaluate: $480 \div (4 \times 3) \div 2 + 5 \times (12 - 2 \times 5)$.

- (A) 90
(C) 60
- (B) 30
(D) 26

40. A reader buys 6 books at ₹p each and 4 magazines at ₹q each, then uses a ₹50 voucher; the bill is also taxed by adding ₹m. Which expression gives the amount paid in ₹?

- (A) $6 \times (p + 4 \times q) - 50 + m$
(C) $6 \times p + 4 \times q - 50 + m$
- (B) $(6 \times p + 4 \times q - 50) \times m$
(D) $6 \times p + 4 \times q - 50 \times m$

41. Evaluate: $7 \times 9 - 3 \times (4 + 2 \times (6 - 1)) + 8$.

- (A) 113
(C) 5
- (B) 47
(D) 29

42. Without fully computing, which expression is the GREATEST?

- (A) $30 \times 45 - 30 \times 2$
(C) $30 \times 40 + 30 \times 3$
- (B) $30 \times 42 + 30$
(D) 30×44

43. Which removal of brackets is CORRECT for $300 - (80 - 40 + 20 - 10 + 5)$?

- (A) $300 - 80 - 40 + 20 - 10 + 5$
(C) $300 - 80 + 40 - 20 + 10 - 5$
- (B) $300 - 80 + 40 + 20 + 10 + 5$
(D) $300 + 80 - 40 + 20 - 10 + 5$

44. To compute 25×44 by the distributive property using $44 = 40 + 4$, the correct working is:

- (A) $25 \times 40 + 4 = 1004$
(C) $25 \times 40 + 25 \times 4 = 1100$
- (B) $25 \times 40 + 25 \times 40 = 2000$
(D) $25 \times 40 - 25 \times 4 = 900$

45. Evaluate: $((48 - 8) \div 5 + 2) \times 3 - 4 \times (7 - 5)$.

- (A) 30
(C) 22
- (B) 14
(D) 34

46. Which statement is TRUE about $96 \div 8 \div 3$ and $96 \div (8 \div 3)$?

- (A) They are equal because division is associative
(B) $96 \div 8 \div 3$ is larger (C) They are equal, both 4
(D) They differ; $96 \div 8 \div 3 = 4$ while $96 \div (8 \div 3) = 36$

47. Without computing, which is greater: 88×88 or 86×90 ?

- (A) 86×90 (B) 88×88
(C) They are equal (D) Cannot be decided

48. Evaluate: $6 + 4 \times (3 + 2 \times (5 - 2 \times 2)) - 7$.

- (A) 19 (B) 43
(C) 27 (D) 11

49. A factory makes 8 machines a day for 15 days, but 2 machines a day fail inspection. Which expression gives the machines that pass, and equals which distributed form?

- (A) $15 \times (8 - 2) = 15 \times 8 - 15 \times 2$ (B) $15 \times 8 - 2 = 15 \times (8 - 2)$
(C) $15 \times (8 - 2) = 15 \times 8 - 2$ (D) $15 \times 8 - 2 \times 8 = 15 \times (8 - 2)$

50. Evaluate: $5 + 3 \times (2 \times (6 - 1) - (12 - 2 \times 5))$.

- (A) 44 (B) 23
(C) 11 (D) 29

51. There are 96 marbles packed 8 to a bag; then 3 full bags are sold and 2 loose marbles are added back. Which expression gives the marbles remaining, and its value?

- (A) $(96 - 3) \times 8 + 2 = 746$ (B) $96 \div 8 - 3 + 2 = 11$
(C) $96 - 3 + 8 + 2 = 103$ (D) $96 - 3 \times 8 + 2 = 74$

52. Written as signed terms, the expression $80 - 5 \times 4 + 60 \div 12 - 2 \times 3$ has which terms?

- (A) 80, -5, 4, 60, 12, -2, 3 (B) 80, -20, 5, -6
(C) 80, 20, 5, 6 (D) 80, -20, -5, -6

53. Evaluate: $4 + 6 \times ((20 - 8) \div (2 + 1)) - 9$.

- (A) 43 (B) 19
(C) 29 (D) 13

54. Without computing, which is greater: $60 \div (12 \div 4)$ or $60 \div 12 \div 4$?

- (A) $60 \div (12 \div 4)$ (B) $60 \div 12 \div 4$
(C) They are equal (D) Cannot be decided

55. Evaluate: $1000 - [400 - (150 - (60 - 25))]$.

- (A) 685 (B) 765
(C) 715 (D) 635

56. Which expression correctly represents 'triple the amount by which 50 exceeds the sum of 12 and 9'?

(A) $3 \times 50 - (12 + 9)$

(B) $3 \times 50 - 12 + 9$

(C) $3 \times (50 - (12 + 9))$

(D) $(3 \times 50 - 12) + 9$

57. Using the distributive property, 99×99 equals:

(A) $99 \times 100 - 99$

(B) $99 \times 100 - 1$

(C) $100 \times 100 - 1$

(D) $99 \times 100 + 99$

58. Evaluate: $7 \times 8 - [40 - 2 \times (15 - 3 \times 4)] + 6$.

(A) 96

(B) 28

(C) 16

(D) 34

59. Which expression has the SAME value as $(72 + 18) \div 9 + 4 \times 3$?

(A) $72 + 18 \div 9 + 12$

(B) $(72 + 18) \div (9 + 4) \times 3$

(C) $90 \div 9 + 12$

(D) $90 \div (9 + 4 \times 3)$

60. Without computing, which is greater: 64×36 or 65×35 ?

(A) 65×35

(B) 64×36

(C) They are equal

(D) Cannot be decided

Solutions — Mathematics (Class 7), Chapter 2: Arithmetic Expressions — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	B	21	A	31	C	41	D	51	D
2	A	12	A	22	A	32	A	42	D	52	B
3	B	13	C	23	D	33	A	43	C	53	B
4	C	14	D	24	C	34	A	44	C	54	A
5	C	15	B	25	B	35	D	45	C	55	C
6	D	16	D	26	A	36	A	46	D	56	C
7	D	17	C	27	C	37	D	47	B	57	A
8	C	18	B	28	B	38	D	48	A	58	B
9	A	19	D	29	D	39	B	49	A	59	C
10	B	20	A	30	B	40	C	50	D	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Innermost first: $2 \times 3 = 6$, so $9 - 6 = 3$. Next bracket: $2 \times 3 = 6$, so $24 - 6 = 18$. Then $5 \times 18 = 90$. Finally $300 - 90 + 7 = 217$. Adding 90 instead of subtracting it gives the too-large value; mishandling the innermost subtraction ($9 - 6 = 15$ by adding) leads to the other inflated results.

2. (A) The inner expression $40 - 15 + 8 - 6 + 3$ carries signs +40, -15, +8, -6, +3. A minus before the bracket reverses each one to -40, +15, -8, +6, -3, giving $250 - 40 + 15 - 8 + 6 - 3$. The wrong options keep one or more signs unflipped.

3. (B) By the distributive property, $84 \times 100 - 84 \times 4 = 84 \times (100 - 4) = 84 \times 96$, the very same product. Picking one side as larger ignores that distribution over a subtraction keeps the value identical.

4. (C) Bracket first: $2 \times 3 = 6$. Then $\times$ and $\div$ left to right: $720 \div 6 = 120$, $120 \div 4 = 30$, $30 \times 5 = 150$. Doing the $\div 4$ and $\times 5$ as $\div (4 \times 5) = \div 20$ wrongly gives 6; multiplying before dividing inflates the result.

5. (C) Each product or quotient collapses to one signed term: $6 \times 4 = 24$ carries its leading minus to -24 ; $72 \div 8 = 9$ stays $+9$; $5 \times 2 = 10$ carries its minus to -10 ; plus the standalone $+100$. So $100, -24, +9, -10$. The split-into-atoms option and the all-positive option both ignore that the $\times$ and $\div$ bind their numbers together with the preceding sign.

6. (D) Factor 17 out of every term: $17 \times (100 - 4 + 1) = 17 \times 97$. Forgetting the final $+17$ (the lone $17 = 17 \times 1$) gives 17×96 ; subtracting it instead gives 17×95 .

7. (D) From the first to the second, the larger number rises by 2 but the amount taken away rises by 3, a net change of $2 - 3 = -1$. So the second difference is 1 less, making $613 - 287$ the greater. Assuming the bigger top means the bigger difference is the trap.

8. (C) Innermost: $7 - 4 = 3$, then $2 \times 3 = 6$, then $12 + 6 = 18$, then $40 - 18 = 22$, then $5 \times 22 = 110$, finally $110 - 18 = 92$. Stopping at 110 (forgetting the -18) or adding instead of subtracting the 18 gives the wrong choices.

9. (A) Three numbers all subtracted may be collected as one subtraction of their sum: $120 - (18 + 24 + 11) = 120 - 53 = 67$, matching $120 - 18 - 24 - 11$. Any inner minus leaves a sign unflipped on removal, and the last option leaves a stray $+11$.

10. (B) $49 \times 51 = (50 - 1)(50 + 1) = 50 \times 50 - 1$, exactly one less than 2500. Splitting a square into two factors that move equally apart always loses, so 50×50 is greater. Believing the spread-out pair stays equal is the misconception.

11. (B) Innermost: $8 - 5 = 3$. Then $36 \div 3 = 12$. Then $144 \div 12 = 12$. Then $6 \times 2 = 12$, and $12 + 12 = 24$. Treating $36 \div (8 - 5)$ as $36 \div 3$ is correct, but dividing 144 by 36 first (ignoring the inner division) gives the wrong small value.

12. (A) Net hourly pay is $95 - 15 = 80$, for 8 hours: $8 \times (95 - 15)$, and the once-only allowance adds 120 outside the hourly product: $8 \times (95 - 15) + 120$. Subtracting only one hour of canteen, or multiplying the daily allowance by 8, mis-models the situation.

13. (C) Remove the inner bracket first: $60 - (25 - 8) = 60 - 25 + 8$. Now $200 - (60 - 25 + 8)$ flips each: $200 - 60 + 25 - 8 = 157$. Two sign-flip stages are needed; flubbing either stage produces the other listed values.

14. (D) Left bracket: $2 + 6 = 8$, so $96 \div 8 = 12$, then $12 \times 5 = 60$. Right bracket: $2 \times 6 = 12$, $15 - 12 = 3$, then $3 \times 3 = 9$. Finally $60 - 9 = 51$. Treating $\div 8 \times 5$ as $\div (8 \times 5) = \div 40$ collapses the first part to 1.2 and derails the rest.

15. (B) $102 = 100 + 2$, so $47 \times 102 = 47 \times 100 + 47 \times 2 = 4700 + 94 = 4794$. Multiplying only the first part and tacking on a bare $+2$ is half-distribution; subtracting the 47×2 treats 102 as $100 - 2$.

16. (D) Pairing $23 + 77 = 100$ and $58 + 42 = 100$, plus the lone 100, gives three hundreds = 300 with only round numbers. The other pairings reach 300 too but never form clean round tens/hundreds, and addition certainly may be regrouped.

17. (C) Both pairs sum to 100. For a fixed sum, the product is larger when the factors are closer together: 38 and 62 differ by 24, while 37 and 63 differ by 26, so 38×62 wins. Assuming equal sums force equal products is the trap.

18. (B) Innermost: $6 + 4 = 10$, then $2 \times 10 = 20$, then $50 - 20 = 30$, then $30 \div 5 = 6$, then $4 \times 6 = 24$, finally $24 + 7 = 31$. Forgetting the $+7$ leaves 24; dividing before subtracting ($50 - 2 \times 10 \div 5$) breaks the bracket structure.

19. (D) Two consecutive subtractions equal subtracting the sum: $60 - 18 - 7 = 60 - (18 + 7) = 35$. The $+7$ version differs, division is not associative so $72 \div 6 \div 2 = 6$ but $72 \div (6 \div 2) = 24$, and $60 - (18 - 7)$ leaves a sign unflipped.

20. (A) Factor out 23: $23 \times (8 + 5 - 3) = 23 \times 10 = 230$. Adding all three coefficients ($8 + 5 + 3 = 16$) gives 368; only adding the first two then ignoring -3 gives $23 \times 13 = 299$ territory; mis-grouping yields the other small values.

21. (A) Products: $9 \times 7 = 63$, $4 \times 6 = 24$, $5 \times 8 = 40$, $3 \times 9 = 27$. With their signs: $63 - 24 + 40 - 27 = 52$. Adding every product as positive gives 154-style errors; flipping the wrong signs gives the negative or the smaller value.

22. (A) $(36 - 12) \div 2 + 4 = 24 \div 2 + 4 = 12 + 4 = 16$. Without brackets the expression is $36 - 6 + 4 = 34$; the option $36 - (12 \div 2 + 4) = 26$, and $36 - 12 \div (2 + 4) = 34$, none of which equal 16.

23. (D) Take the first sum minus the second, term by term: $(124 - 126) + (89 - 88) + (211 - 209) = -2 + 1 + 2 = +1$. The total is positive, so the first sum is 1 greater. The trap is stopping after the first column, where the second sum looks ahead.

24. (C) Innermost: $10 - 4 = 6$, then $2 \times 6 = 12$, then $18 + 12 = 30$, then $50 - 30 = 20$, then $3 \times 20 = 60$, finally $250 - 60 = 190$. Mishandling the deepest bracket or the final subtraction gives the distractors.

25. (B) Inner first: $8 - 3 = 5$, so $20 - 5 = 15$, then $90 - 15 = 75$. Equivalently remove inner: $20 - (8 - 3) = 20 - 8 + 3$, then $90 - (20 - 8 + 3) = 90 - 20 + 8 - 3 = 75$. Dropping all minus-flips gives 59; flipping inconsistently gives 81 or 65.

26. (A) $98 = 100 - 2$, so $26 \times 98 = 26 \times 100 - 26 \times 2 = 2600 - 52 = 2548$. Subtracting a bare 2 is half-distribution; $90 + 8$ is not a valid factor split that keeps 26 attached; adding 26×2 treats 98 as 102.

27. (C) A minus before a bracket flips both inner signs, a plus keeps them. So $-(25 - 12) \rightarrow -25 + 12$, $+(18 - 9) \rightarrow +18 - 9$, $-(6 - 2) \rightarrow -6 + 2$. Chained: $80 - 25 + 12 + 18 - 9 - 6 + 2$. Each wrong option leaves one sign unflipped.

28. (B) Collapse products and quotients ($\times$ and $\div$ left to right): $60 \div 5 = 12$; $7 \times 3 = 21$; $24 \div 8 = 3$, then $\times 2 = 6$. So $100 - 12 + 21 - 6 = 103$. Doing $24 \div (8 \times 2) = 1.5$ or adding the 12 instead of subtracting gives the wrong totals.

29. (D) Both factor-pairs sum to 100. Closer factors give the larger product: 53 and 47 differ by 6, while 54 and 46 differ by 8, so 53×47 is greater. Equal sums do not force equal products.

30. (B) Each box yields $8 - 3 = 5$ good ones, across 12 boxes: $12 \times (8 - 3) = 12 \times 5 = 60$ (equivalently $12 \times 8 - 12 \times 3 = 96 - 36 = 60$). Subtracting only 3 once, or reducing the box count, mis-models the per-box damage.

31. (C) Compute each: $(3 + 5)(4 - 1) = 8 \times 3 = 24$; $3 + 5 \times 4 - 1 = 3 + 20 - 1 = 22$; $3 \times 5 + 4 \times 1 = 15 + 4 = 19$; $3 \times 5 - 4 + 1 = 12$. The bracketed product 24 is largest; left-to-right slips would wrongly inflate the second option.

32. (A) Work from the deepest: $8 - 3 = 5$; $20 - 5 = 15$; $50 - 15 = 35$; $60 - 35 = 25$. Each nested subtraction must be finished before the layer outside it; skipping a layer yields 15, 35, or 45.

33. (A) Factor 34: $34 \times (45 + 55 - 100) = 34 \times 0 = 0$. Since $45 + 55 = 100$, the bracket vanishes. Forgetting to subtract the 100, or adding it, leaves a nonzero multiple of 34.

34. (A) Inner signs are +50, -20, +10; the leading minus flips each to -50, +20, -10, giving $180 - 50 + 20 - 10$. Every wrong option leaves at least one inner sign unflipped.

35. (D) $7 \times (30 - 5) = 7 \times 30 - 7 \times 5$ takes away 35, while $7 \times 30 - 5$ takes away only 5, so $7 \times 30 - 5$ is larger. Forgetting that distribution multiplies the subtracted 5 by 7 leads to calling them equal.

36. (A) Left: $5 - 1 = 4$, $2 \times 4 = 8$, $8 + 8 = 16$, $3 \times 16 = 48$. Right: $4 - 2 = 2$, $3 \times 2 = 6$, $10 - 6 = 4$, $2 \times 4 = 8$. Then $48 - 8 = 40$. Adding the two parts gives 56; mishandling a nested subtraction shifts the result to 24 or 16.

37. (D) Each team has $9 + 2 = 11$ people, five teams: $5 \times (9 + 2) = 55$; only 3 players total leave, subtracted once: $5 \times (9 + 2) - 3 = 52$. Subtracting 3 inside the bracket removes 3 per team (15 total), which is wrong.

38. (D) Common factor 19: $19 \times (23 + 17) = 19 \times 40$. The half-distributed $19 \times 23 + 17$ drops a factor of 19 from the second term; doubling 19 to 38 double-counts the multiplier; 19×391 multiplies the addends instead of adding them.

39. (B) Bracket $4 \times 3 = 12$, then $480 \div 12 = 40$, then $40 \div 2 = 20$. Right: $2 \times 5 = 10$, $12 - 10 = 2$, $5 \times 2 = 10$. So $20 + 10 = 30$. Doing $\div 12$ then $\times$ the right block before the second division, or $480 \div 12 \div 2$ as $480 \div 6$, throws off the total.

40. (C) Book cost $6 \times p$, magazine cost $4 \times q$ add together; the voucher subtracts 50 once and the tax adds m once: $6 \times p + 4 \times q - 50 + m$. Wrapping the magazine cost into the book bracket, multiplying the whole bill by m , or multiplying 50 by m all distort the model.

41. (D) First $7 \times 9 = 63$. Bracket: $6 - 1 = 5$, $2 \times 5 = 10$, $4 + 10 = 14$, $3 \times 14 = 42$. So $63 - 42 + 8 = 29$. Adding 42 instead of subtracting gives 113; dropping the $+8$ gives 21-style errors near $47/5$ from misordered steps.

42. (D) Rewrite each as $30 \times$ something: $30 \times 45 - 30 \times 2 = 30 \times 43$; $30 \times 42 + 30 = 30 \times 43$; $30 \times 40 + 30 \times 3 = 30 \times 43$; and the lone $30 \times 44 = 30 \times 44$. Since $44 > 43$, 30×44 is greatest. Each distractor was engineered to collapse to 30×43 .

43. (C) Inner signs $+80$, -40 , $+20$, -10 , $+5$ each flip under the leading minus: -80 , $+40$, -20 , $+10$, -5 , giving $300 - 80 + 40 - 20 + 10 - 5$. The wrong options keep one or more original signs.

44. (C) $25 \times 44 = 25 \times (40 + 4) = 25 \times 40 + 25 \times 4 = 1000 + 100 = 1100$. Adding a bare 4 is half-distribution; subtracting 25×4 treats 44 as 36; doubling 25×40 misreads the split.

45. (C) Innermost: $48 - 8 = 40$, $40 \div 5 = 8$, $8 + 2 = 10$, $10 \times 3 = 30$. Right: $7 - 5 = 2$, $4 \times 2 = 8$. Then $30 - 8 = 22$. Forgetting the right block leaves 30; mishandling the division order gives the smaller values.

46. (D) Left to right: $96 \div 8 = 12$, then $\div 3 = 4$. With the bracket: $8 \div 3$ first, then $96 \div (8 \div 3) = 96 \times 3 \div 8 = 36$. Division is not associative, so the bracket genuinely changes the value; the two are not equal.

47. (B) Both pairs have the same sum 176 and centre 88. The square 88×88 has factors at zero spread; 86×90 spreads them by 4 and equals $88 \times 88 - 4$. For a fixed sum the product shrinks as the factors part, so 88×88 is greater.

48. (A) Innermost: $2 \times 2 = 4$, $5 - 4 = 1$, $2 \times 1 = 2$, $3 + 2 = 5$, $4 \times 5 = 20$, then $6 + 20 - 7 = 19$. Reading $5 - 2 \times 2$ as $(5 - 2) \times 2 = 6$ derails the nest into 43; dropping the -7 gives 26-ish errors.

49. (A) Each day yields $8 - 2 = 6$ good machines, over 15 days: $15 \times (8 - 2)$, and distribution gives $15 \times 8 - 15 \times 2 = 120 - 30 = 90$. The other options either subtract only 2 total or distribute incompletely, so the stated equality fails for them.

50. (D) Inner left: $6 - 1 = 5$, $2 \times 5 = 10$. Inner right: $2 \times 5 = 10$, $12 - 10 = 2$. Bracket: $10 - 2 = 8$, $3 \times 8 = 24$, then $5 + 24 = 29$. Adding the inner pieces instead of subtracting, or

dropping the +5, gives the wrong values.

51. (D) Three sold bags remove $3 \times 8 = 24$ marbles, and 2 are added back: $96 - 3 \times 8 + 2 = 96 - 24 + 2 = 74$. Bracketing $96 - 3$ first, or counting bags instead of marbles, mis-models the loss.

52. (B) Collapse: $5 \times 4 = 20$ with its minus $\rightarrow -20$; $60 \div 12 = 5 \rightarrow +5$; $2 \times 3 = 6$ with its minus $\rightarrow -6$; plus +80. So 80, -20, +5, -6. The atomised list and the all-positive list both ignore that $\times/\div$ bind to their preceding sign.

53. (B) Brackets: $20 - 8 = 12$, $2 + 1 = 3$, $12 \div 3 = 4$. Then $6 \times 4 = 24$, and $4 + 24 - 9 = 19$. Reading $(20 - 8) \div (2 + 1)$ as $20 - 8 \div 2 + 1$ breaks the structure; dropping the -9 gives 28.

54. (A) $60 \div (12 \div 4) = 60 \div 3 = 20$, while $60 \div 12 \div 4 = 5 \div 4 = 1.25$. Dividing by a smaller number gives a larger result, so the bracketed form is far greater. Treating division as associative makes them look equal.

55. (C) Deepest: $60 - 25 = 35$; $150 - 35 = 115$; $400 - 115 = 285$; $1000 - 285 = 715$. Each layer must close before the one outside it; skipping or mis-signing a layer yields the other values.

56. (C) The amount by which 50 exceeds $12 + 9$ is $50 - (12 + 9) = 29$; tripling it needs the whole difference inside a bracket: $3 \times (50 - (12 + 9)) = 3 \times 29 = 87$. Tripling only the 50 first, or losing the sum-bracket, changes the meaning.

57. (A) $99 \times 99 = 99 \times (100 - 1) = 99 \times 100 - 99 \times 1 = 9900 - 99 = 9801$. Subtracting a bare 1 is half-distribution; $100 \times 100 - 1$ changes the factor entirely; adding 99 treats the multiplier as 101.

58. (B) $7 \times 8 = 56$. Inside: $3 \times 4 = 12$, $15 - 12 = 3$, $2 \times 3 = 6$, $40 - 6 = 34$. Then $56 - 34 + 6 = 28$. Adding 34 instead of subtracting, or mishandling the bracket sign, produces 96 and the smaller wrong values.

59. (C) Brackets first: $72 + 18 = 90$, so the expression is $90 \div 9 + 4 \times 3 = 10 + 12 = 22$, matching $90 \div 9 + 12$. The first distractor divides only 18 by 9; the others fuse the addition into the divisor, changing the value.

60. (B) Both pairs sum to 100. The pair closer together has the larger product: 64 and 36 differ by 28, while 65 and 35 differ by 30, so 64×36 is greater. Equal sums do not make the products equal.